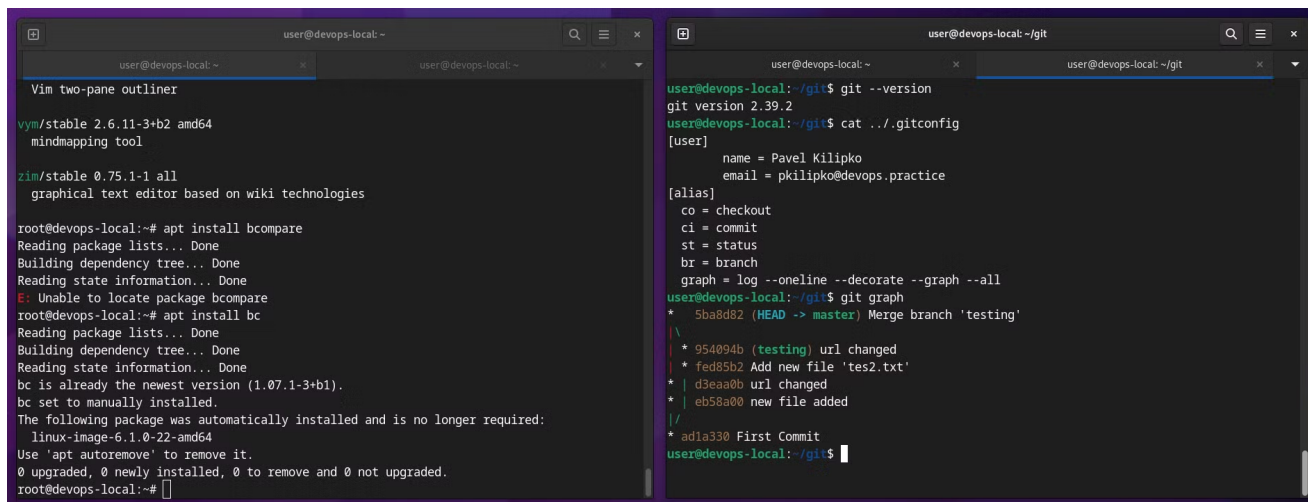


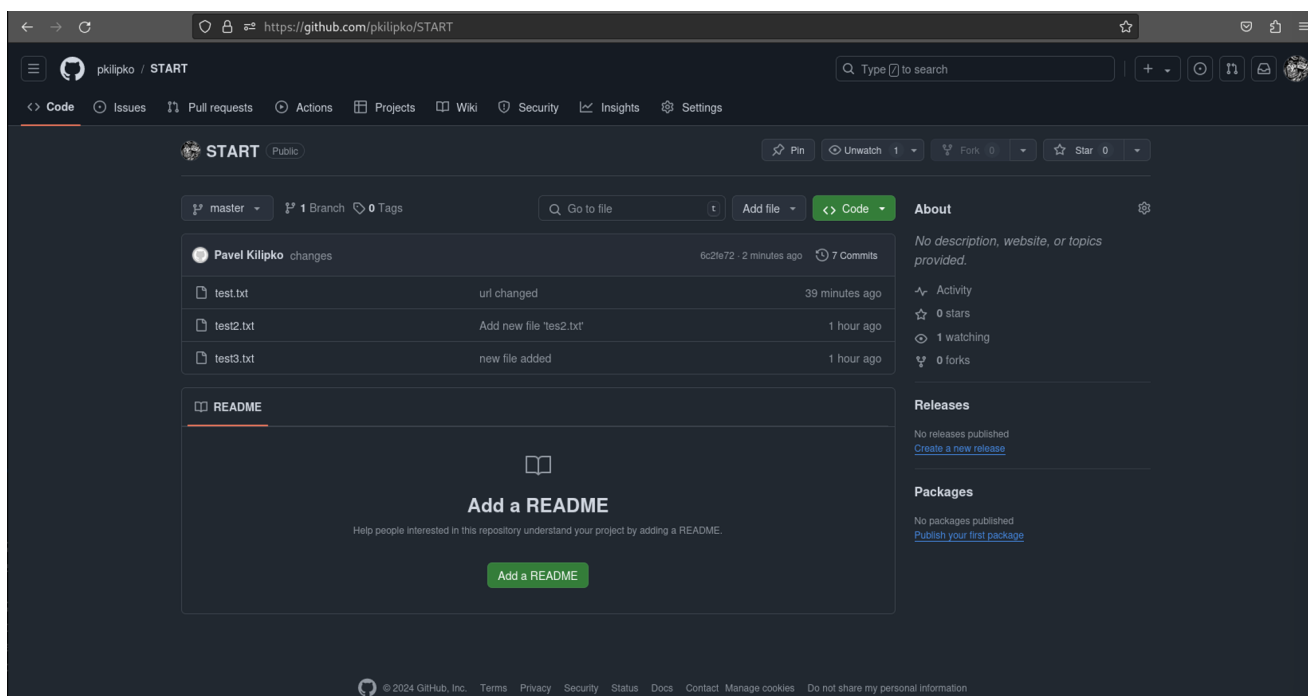
Практические навыки из лекционного материала

Система контроля версий

Начальная VM с установленным git и результаты практики:



```
user@devops-local: ~  
Vim two-pane outliner  
vym/stable 2.6.11-3+b2 amd64  
  mindmapping tool  
zim/stable 0.75.1-1 all  
  graphical text editor based on wiki technologies  
root@devops-local:~# apt install bcompare  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
E: Unable to locate package bcompare  
root@devops-local:~# apt install bc  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
bc is already the newest version (1.07.1-3+b1).  
bc set to manually installed.  
The following package was automatically installed and is no longer required:  
  linux-image-6.1.0-22-amd64  
Use 'apt autoremove' to remove it.  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
root@devops-local:~#  
user@devops-local: ~/git  
user@devops-local:~/git$ git --version  
git version 2.39.2  
user@devops-local:~/git$ cat ../.gitconfig  
[user]  
  name = Pavel Kilipko  
  email = pkilipko@devops.practice  
[alias]  
  co = checkout  
  ci = commit  
  st = status  
  br = branch  
  graph = log --online --decorate --graph --all  
user@devops-local:~/git$ git graph  
* 5ba8d82 (HEAD -> master) Merge branch 'testing'  
|  
* 0954094b (testing) url changed  
* fed85b2 Add new file 'tes2.txt'  
* | d3eaa0b url changed  
* | eb58a00 new file added  
|/  
* ad1a330 First Commit  
user@devops-local:~/git$
```



GitLab сервер и GitLab Runner

Для раннеров две отдельные VM'ки:

```
user@devops-local-2: ~  
root@devops-local-2:~# neofetch  
_,met$$$$$ggg.      root@devops-local-2  
,g$$$$$$$$$$$$$SP.  
,"Y$$$.  
,$$P'      $$$.  
',$$P'      ,ggs.      $Sb:  
'd$$'      ,SP"      $$$  
,$$P      d$'      $SP  
$$:      $$      ,d$$'  
$$;      Y$b._      ,d$P'  
Y$$      .,"Y$$$$$P"  
`$$b      "-._  
`Y$$  
`$$b.  
`Y$$b.  
`Y$b._  
`"  
root@devops-local-2:~#  
OS: Debian GNU/Linux 12 (bookworm) x86_64  
Host: VMware Virtual Platform None  
Kernel: 6.1.0-23-amd64  
Uptime: 8 hours, 17 mins  
Packages: 1713 (dpkg)  
Shell: bash 5.2.15  
Resolution: 1280x800  
CPU: 12th Gen Intel i5-1240P (8) @ 2.111GHz  
GPU: 00:0f.0 VMware SVGA II Adapter  
Memory: 494MiB / 1932MiB
```

```
user@devops-local-3: ~  
root@devops-local-3:~# neofetch  
_,met$$$$$ggg.      root@devops-local-3  
,g$$$$$$$$$$$$$SP.  
,"Y$$$.  
,$$P'      $$$.  
',$$P'      ,ggs.      $Sb:  
'd$$'      ,SP"      $$$  
,$$P      d$'      $SP  
$$:      $$      ,d$$'  
$$;      Y$b._      ,d$P'  
Y$$      .,"Y$$$$$P"  
`$$b      "-._  
`Y$$  
`$$b.  
`Y$$b.  
`Y$b._  
`"  
root@devops-local-3:~#  
OS: Debian GNU/Linux 12 (bookworm) x86_64  
Host: VMware Virtual Platform None  
Kernel: 6.1.0-23-amd64  
Uptime: 1 min  
Packages: 1722 (dpkg)  
Shell: bash 5.2.15  
Resolution: 1280x800  
CPU: 12th Gen Intel i5-1240P (8) @ 2.112GHz  
GPU: 00:0f.0 VMware SVGA II Adapter  
Memory: 401MiB / 1932MiB
```

Запускаем на них два раннера:

The image displays the GitLab CI/CD settings interface and two terminal windows showing the startup of GitLab runners.

GitLab CI/CD Settings:

- General pipelines:** Customize your pipeline configuration.
- Auto DevOps:** Automate building, testing, and deploying your applications based on your continuous integration and delivery configuration. [How do I get started?](#)
- Runners:** Runners are processes that pick up and execute CI/CD jobs for GitLab. [What is GitLab Runner?](#)
Register as many runners as you want. You can register runners as separate users, on separate servers, and on your local machine.
How do runners pick up jobs?
Runners are either:
 - active** - Available to run jobs.
 - paused** - Not available to run jobs.Tags control which type of jobs a runner can handle. By tagging a runner, you make sure runners only handle the jobs they are equipped to run. [Learn more.](#)
- Project runners:** These runners are assigned to this project. [New project runner](#)
- Instance runners:** These runners are available to all groups and projects.
Enable instance runners for this project: ☒
This GitLab instance does not provide any instance runners yet. Administrators can register instance runners in the admin area.
- Group runners:** These runners are shared across projects in this group.
Group runners can be managed with the [Runner API](#).
This project does not belong to a group and cannot make use of group runners.
- Artifacts:** A job artifact is an archive of files and directories saved by a job when it finishes.
- Variables:** Variables store information that you can use in job scripts. Each project can define a maximum of 8000 variables. [Learn more.](#)
- Pipeline trigger tokens:** Trigger a pipeline for a branch or tag by generating a trigger token and using it with an API call. The token impersonates a user's project access and permissions. [Learn more.](#)
- Deploy freezes:** Add a freeze period to prevent unintended releases during a period of time for a given environment. You must update the deployment jobs in `.gitlab-ci.yml` according to the deploy freezes added here. [Learn more.](#) Specify deploy freezes using [cron syntax](#).

Terminal Windows:

Terminal 1 (devops-local-2):

```
root@devops-local-2:~# gitlab-runner run
Runtime platform arch=amd64 os=linux pid=6910
revision=66269445 version=17.3.1
Starting multi-runner from /etc/gitlab-runner/config.toml... builds=0 max_builds=0
Running in system-mode.

Configuration loaded builds=0 max_builds=1
listen_address not defined, metrics & debug endpoints disabled builds=0 max_builds=1
[session_server].listen_address not defined, session endpoints disabled builds=0 max_builds=1
Initializing executor providers builds=0 max_builds=1
[]
```

Terminal 2 (devops-local-3):

```
root@devops-local-3:~# gitlab-runner run
Runtime platform arch=amd64 os=linux pid=1405
revision=66269445 version=17.3.1
Starting multi-runner from /etc/gitlab-runner/config.toml... builds=0 max_builds=0
Running in system-mode.

Configuration loaded builds=0 max_builds=1
listen_address not defined, metrics & debug endpoints disabled builds=0 max_builds=1
[session_server].listen_address not defined, session endpoints disabled builds=0 max_builds=1
Initializing executor providers builds=0 max_builds=1
[]
```

Создаем тестовые файлы и конфиг пайпа. На одном из раннеров видим развернутый сайт:

The image shows a GitLab Pipelines interface in a web browser. The browser tabs include 'pkilipko/START' and 'Pipelines - Administrator'. The address bar shows 'gitlab.local/root/start/-/pipelines'. The left sidebar contains a 'Project' menu with options like 'START', 'Pinned', 'Issues', 'Merge requests', 'Manage', 'Plan', 'Code', 'Build', 'Pipelines', 'Jobs', 'Pipeline editor', 'Pipeline schedules', 'Artifacts', 'Secure', 'Deploy', 'Operate', 'Monitor', 'Analyze', and 'Settings'. The main content area displays a table of pipelines with columns for 'Status', 'Pipeline', 'Created by', and 'Stages'. A single pipeline is shown with a 'Passed' status, named 'Add new file', and a 'latest' tag. Below the table, a preview of the pipeline's output is shown, displaying 'hello world!' in a browser window.

Below the main interface, two terminal windows are shown. The left terminal, titled 'user@devops-local-2: ~', displays logs for a GitLab runner job. The right terminal, titled 'user@devops-local-3: ~', displays the output of the 'gitlab-runner run' command.

```
user@devops-local-2: ~  
WARNING: Checking for jobs... failed runner=ZctmFgFjH status=502  
Bad Gateway  
Checking for jobs... received job=38 repo_url=http://gitlab.local/root/start.git  
Added job to processing list builds=1 job=38 max_builds=1  
project=1 repo_url=http://gitlab.local/root/start.git time_in_queue_seconds=2  
Appending trace to coordinator...ok code=202 job=38 job-log=0-68  
7 job-status=running runner=ZctmFgFjH sent-log=0-686 status=202 Accepted update-interval=1m0s  
Appending trace to coordinator...ok code=202 job=38 job-log=0-50  
31 job-status=running runner=ZctmFgFjH sent-log=687-5030 status=202 Accepted update-interval=1m0s  
Job succeeded duration_s=77.55938621 job=38  
8 project=1 runner=ZctmFgFjH  
Appending trace to coordinator...ok code=202 job=38 job-log=0-51  
54 job-status=running runner=ZctmFgFjH sent-log=5031-5153 status=202 Accepted update-interval=3s  
Updating job... bytesize=5154 checksum=crc32  
:ab788d05 job=38 runner=ZctmFgFjH  
Submitting job to coordinator...ok bytesize=5154 checksum=crc32  
:ab788d05 code=200 job=38 job-status=success runner=ZctmFgFjH update-interval=0s  
Removed job from processing list builds=0 job=38 max_builds=1  
project=1 repo_url=http://gitlab.local/root/start.git time_in_queue_seconds=2
```

```
user@devops-local-3: ~  
root@devops-local-3:~# gitlab-runner run  
Runtime platform arch=amd64 os=linux pid=2421  
revision=66269445 version=17.3.1  
Starting multi-runner from /etc/gitlab-runner/config.toml... builds=0 max_builds=0  
Running in system-mode.  
  
Configuration loaded builds=0 max_builds=1  
listen_address not defined, metrics & debug endpoints disabled builds=0 max_builds=1  
[session_server].listen_address not defined, session endpoints disabled builds=0 max_builds=1  
Initializing executor providers builds=0 max_builds=1
```

Ansible

Ставим ansible, настраиваем inventory и проверяем работу модулей:

The image shows two terminal windows. The left window shows the execution of the command `ansible host01 -m ping`. The output indicates that the discovered Python interpreter at `/usr/bin/python3.11` is being used. The right window shows the output of the command, displaying a JSON object with various system facts for host01, including uptime, user information, and virtualization details.

```
root@devops-local:~# ansible host01 -m ping
[WARNING]: Platform linux on host host01 is using the discovered Python
interpreter at /usr/bin/python3.11, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
host01 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.11"
  },
  "changed": false,
  "ping": "pong"
}
root@devops-local:~#
```

```
"ansible_uptime_seconds": 2789,
"ansible_user_dir": "/home/ansible",
"ansible_user_gecos": ",,,",
"ansible_user_gid": 1003,
"ansible_user_id": "ansible",
"ansible_user_shell": "/bin/bash",
"ansible_user_uid": 1003,
"ansible_userspace_architecture": "x86_64",
"ansible_userspace_bits": "64",
"ansible_virtualization_role": "guest",
"ansible_virtualization_tech_guest": [
  "VMware"
],
"ansible_virtualization_tech_host": [],
"ansible_virtualization_type": "VMware",
"discovered_interpreter_python": "/usr/bin/python3.11",
"gather_subset": [
  "all"
],
"module_setup": true
},
"changed": false
}
root@devops-local:~#
```

The image shows a sequence of three terminal windows. The first window shows the creation of a file `test-file.txt` in the `/tmp` directory. The second window shows the editing of the `ansible.cfg` file. The third window shows the execution of the command `ansible host01 -m file -a "path=/tmp/test-file.txt state=touch"`, which results in the file being created on the remote host. The output of the command is shown in the third window.

```
test-file.txt
0 Sep 5 19:41
test-file.txt
77G / 97G (79%)
ansible.cfg
77G / 97G (79%)
Hint: If you want to see your .* files, say so in the Configuration dialog.
root@devops-local:~#
```

```
1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit
root@devops-local:~#
```

```
host01 | CHANGED | rc=0 >>
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:61:17:26 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.11.130/24 brd 192.168.11.255 scope global dynamic noprefixroute ens33
        valid_lft 1013sec preferred_lft 1013sec
    inet6 fe80::20c:29ff:fe61:1726/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: outline-tun0: <NO-CARRIER,POINTOPOINT,MULTICAST,NOARP,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 500
    link/nop
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:4f:13:50:de brd ff:ff:ff:ff:ff:ff
root@devops-local:~#
```

```
"changed": false
}
root@devops-local:~# ansible host01 -m file -a "path=/tmp/test-file.txt state=touch"
[WARNING]: Platform linux on host host01 is using the discovered Python
interpreter at /usr/bin/python3.11, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
host01 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.11"
  },
  "changed": true,
  "dest": "/tmp/test-file.txt",
  "gid": 1003,
  "group": "ansible",
  "mode": "0644",
  "owner": "ansible",
  "size": 0,
  "state": "file",
  "uid": 1003
}
root@devops-local:~#
```

Запускаем плейбук:

```
user@devops-local: ~  
root@devops-local:~# cat test-playbook.yaml  
- name: Test  
  hosts: all # playbook запустится для всех хостов в inventory.  
  become: yes # команда позволяет ansible использовать права root.  
  
  tasks: # ниже начинается список задач. Обратите внимание, tasks во мно  
          жественном числе. Иначе получим ошибку при выполнении плейбука  
  
    - name: Create Groups  
      group:  
        name: "{{item}}" # параметр item по сути является переменной. В lo  
                          ор для нее описываются значения, которые она примет.  
        state: present  
      loop: # в результате мы создадим две группы с названиями  
            и dev и test.  
        - dev  
        - test  
  
    - name: Create User  
      user:  
        name: "{{item.client_name}}" # похожим образом item работает и здесь.  
        в with_items указываются значения, которые они должны принять.  
        shell: /bin/bash/ # какую командную оболочку предоставлять  
        юзеру.
```

```
user@devops-local: ~  
TASK [Gathering Facts] *****  
[WARNING]: Platform linux on host host01 is using the discovered Python  
interpreter at /usr/bin/python3.11, but future installation of another Python  
interpreter could change the meaning of that path. See  
https://docs.ansible.com/ansible-  
core/2.17/reference_appendices/interpreter_discovery.html for more information.  
ok: [host01]  
  
TASK [Create Groups] *****  
changed: [host01] => (item=dev)  
changed: [host01] => (item=test)  
  
TASK [Create User] *****  
changed: [host01] => (item={'client_name': 'test', 'home_dir': 'home_dir_test'})  
  
TASK [Installing package] *****  
changed: [host01]  
  
PLAY RECAP *****  
host01 : ok=4 changed=3 unreachable=0 failed=0 s  
kipped=0 rescued=0 ignored=0  
root@devops-local:~#
```

Оставляем на основном хосте работающий сервис gitlab'a. На одном из раннеров запускаем compose-файл с мониторингом, указав в параметрах адрес хоста с gitlab:

File Edit View History Bookmarks Tools Help

Administrator / STA x C/CD Settings - C/ x GitLab CI pipelines x Problem loading po x Prometheus Time S x Prometheus Time S x localhost:9090/me x GitLab CI pipelines x

192.168.11.137:3000/d/gitlab_ci_pipelines/gitlab-ci-pipelines?orgId=1&refresh=10s

General / GitLab CI pipelines ☆ 🔔

owner All project All ref All

PIPELINES... 1 FAILED PL... 0 RUNS # (in the last h... 0

Average Pipeline Run Frequency N/A Average Pipeline Duration 0 seconds

PIPELINE RUNS

1 0.500 0 22:40 22:50 23:00 23:10 23:20 23:30

RUNNING, FAILED OR NOT COMPLETED

SUCCESSFULLY COMPLETED

ID	Project	Ref Kind	Ref Name	Date ↕	Durat
2	root/start	branch	main	34 minutes ago	0 sec

```
user@devops-local: ~  
• gitlab-runservdir.service - GitLab Runit supervision process  
Loaded: loaded (/lib/systemd/system/gitlab-runservdir.service; enabled; pres  
Active: active (running) since Thu 2024-09-05 21:50:27 MSK; 1h 44min ago  
Main PID: 6448 (runsrvdir)  
Tasks: 452 (limit: 4915)  
Memory: 4.0G  
CPU: 52min 43.193s  
CGroup: /system.slice/gitlab-runservdir.service  
├─ 6448 runsrvdir -P /opt/gitlab/service "log: .....  
├─ 6449 runsrv postgresql  
├─ 6450 runsrv puma  
├─ 6451 runsrv prometheus  
├─ 6452 runsrv redis  
├─ 6453 runsrv sidekiq  
├─ 6454 runsrv nginx  
├─ 6455 runsrv gitlab-exporter  
├─ 6456 runsrv redis-exporter  
├─ 6457 runsrv gitlab-workhorse  
├─ 6458 runsrv alertmanager  
├─ 6459 runsrv logrotate  
├─ 6460 runsrv gitaly  
├─ 6461 runsrv node-exporter  
└─ 6462 runsrv gitlab-kas  
lines 1-23
```

```
user@devops-local-3: ~/GitlabCI-exporter  
user@devops-local-3:~/GitlabCI-exporter$ sudo docker-compose ps  
Name Command State Ports  
-----  
gitlabci- /usr/local/bin/gitlab Up 0.0.0.0:8080-  
exporter_gitlab-ci- -ci-p ... >8080/tcp,:::8080-  
pipelines-exporter_1 >8080/tcp  
grafana /run.sh Up 0.0.0.0:3000-  
>3000/tcp,:::3000-  
prometheus /bin/prometheus Up 0.0.0.0:9090-  
--config.f ... >9090/tcp,:::9090-  
>9090/tcp  
user@devops-local-3:~/GitlabCI-exporter$
```

Задание к блоку №4. GITHUB

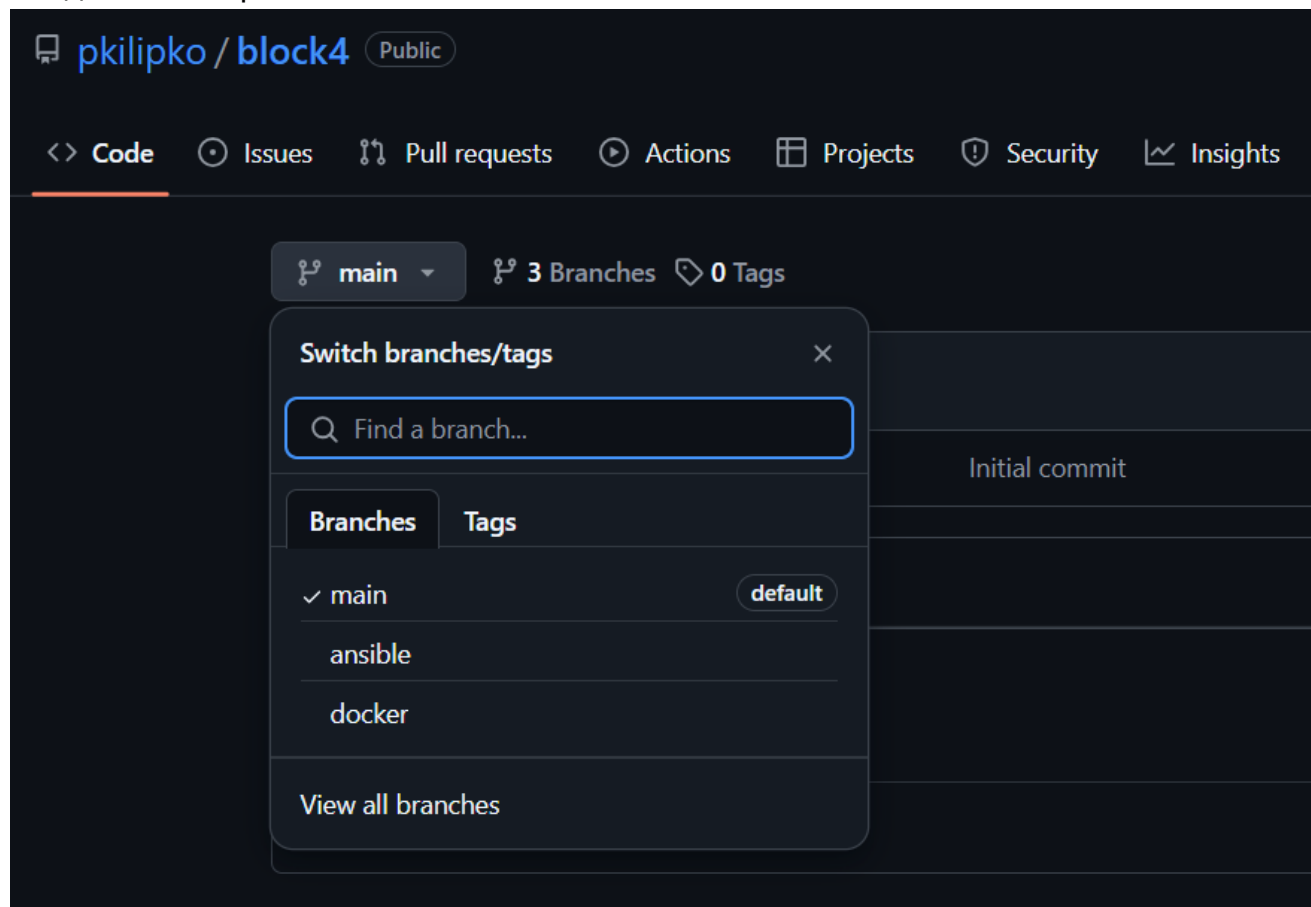
Ветка docker

Репы:

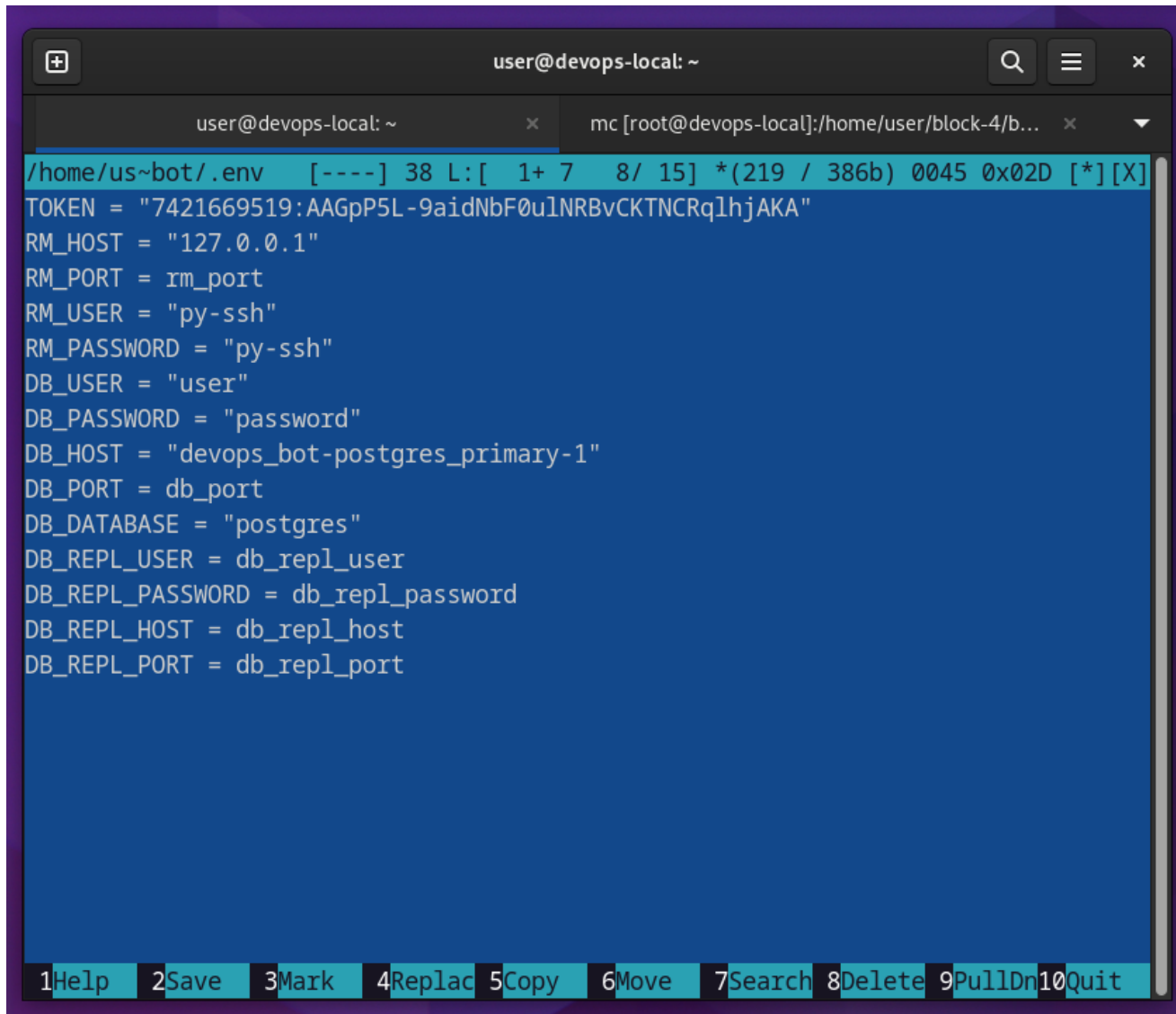
<https://github.com/pkilipko/START> - учебно-практическая репа;

<https://github.com/pkilipko/block4> - задание блока.

Созданы ветки репы:



Прописываем переменные в .env файл:

A screenshot of a terminal window with a dark background. The window title is "user@devops-local: ~". The terminal shows the contents of a file named ".env" located at "/home/user/bot/". The file contains several environment variables for a Redis Master (RM) and a PostgreSQL database (DB). The variables include TOKEN, RM_HOST, RM_PORT, RM_USER, RM_PASSWORD, DB_USER, DB_PASSWORD, DB_HOST, DB_PORT, DB_DATABASE, DB_REPL_USER, DB_REPL_PASSWORD, DB_REPL_HOST, and DB_REPL_PORT. The terminal also shows a status bar at the bottom with various keyboard shortcuts like 1Help, 2Save, 3Mark, etc.

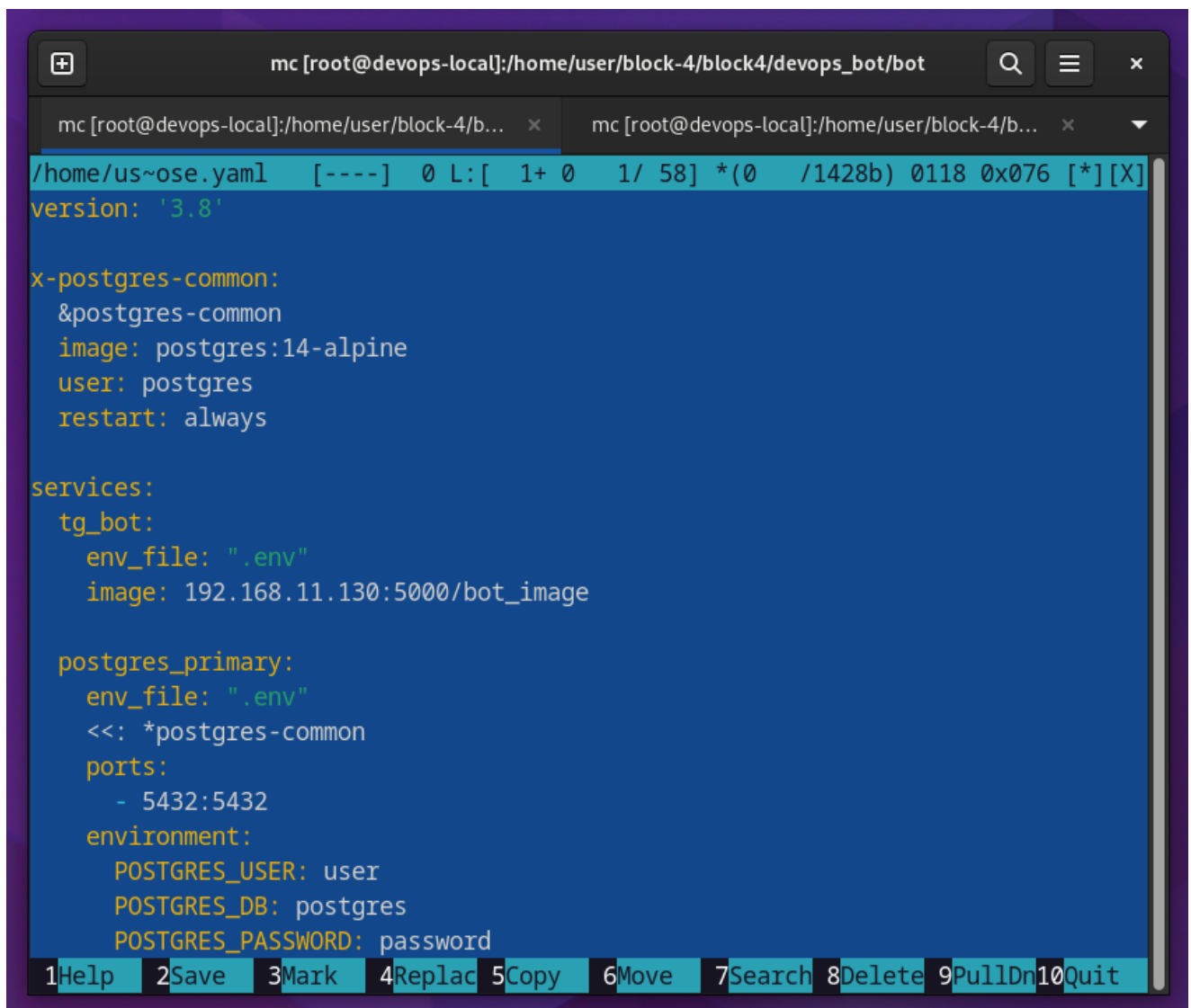
```
user@devops-local: ~
/home/user/bot/.env  [----] 38 L: [ 1+ 7 8/ 15] *(219 / 386b) 0045 0x02D [*] [X]
TOKEN = "7421669519:AAGpP5L-9aidNbF0u1NRBvCKTNCrqlhjAKA"
RM_HOST = "127.0.0.1"
RM_PORT = rm_port
RM_USER = "py-ssh"
RM_PASSWORD = "py-ssh"
DB_USER = "user"
DB_PASSWORD = "password"
DB_HOST = "devops_bot-postgres_primary-1"
DB_PORT = db_port
DB_DATABASE = "postgres"
DB_REPL_USER = db_repl_user
DB_REPL_PASSWORD = db_repl_password
DB_REPL_HOST = db_repl_host
DB_REPL_PORT = db_repl_port

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn10Quit
```

В docker-compose файле добавляем использование файла .env:

⚠ Что можно улучшить.

В сервисы пробрасываются все переменные. Можно расписать переменные по сервисам, а не пробрасывать весь набор туда, где это ненужно.



```
mc [root@devops-local]:/home/user/block-4/block4/devops_bot/bot
/home/us~ose.yaml [----] 0 L: [ 1+ 0 1/ 58] *(0 /1428b) 0118 0x076 [*][X]
version: '3.8'

x-postgres-common:
  &postgres-common
  image: postgres:14-alpine
  user: postgres
  restart: always

services:
  tg_bot:
    env_file: ".env"
    image: 192.168.11.130:5000/bot_image

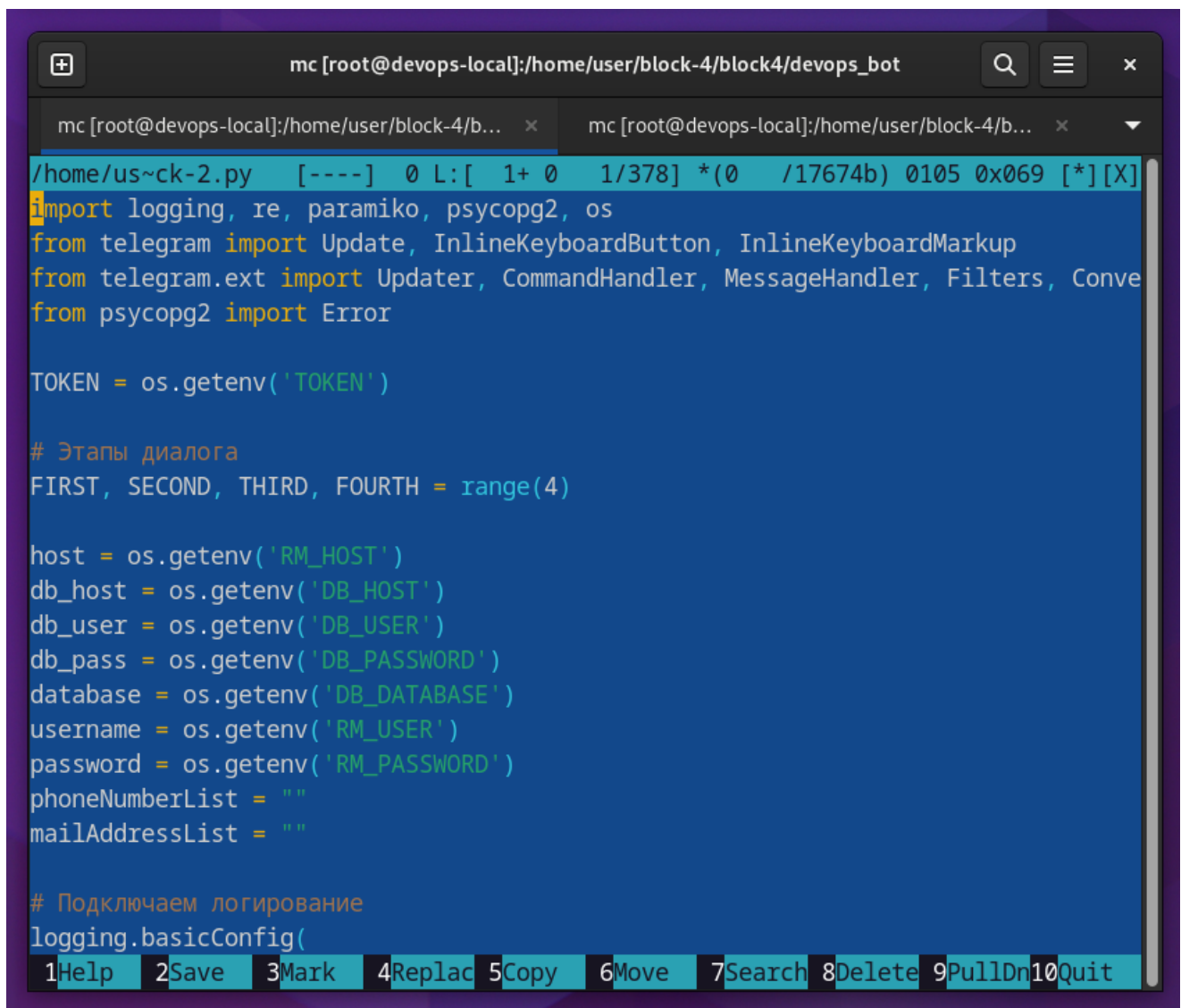
  postgres_primary:
    env_file: ".env"
    <<: *postgres-common
    ports:
      - 5432:5432
    environment:
      POSTGRES_USER: user
      POSTGRES_DB: postgres
      POSTGRES_PASSWORD: password
```

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn10Quit

Обновляем скрипт бота - добавляем модуль получения переменных окружения, которые будут проброшены compose из .env и задаем переменные:

Переменные gitlab.

В gitlab есть встроенное хранилище переменных, которые могут быть безопасно проброшены в пайплайн на этапе выполнения джоб. Более сложным и функциональным является использование внешних хранилищ наподобие Hashicorp Vault.



```
mc [root@devops-local]:/home/user/block-4/block4/devops_bot
/home/us~ck-2.py [----] 0 L: [ 1+ 0 1/378] *(0 /17674b) 0105 0x069 [*][X]
import logging, re, paramiko, psycpg2, os
from telegram import Update, InlineKeyboardButton, InlineKeyboardMarkup
from telegram.ext import Updater, CommandHandler, MessageHandler, Filters, Conve
from psycpg2 import Error

TOKEN = os.getenv('TOKEN')

# Этапы диалога
FIRST, SECOND, THIRD, FOURTH = range(4)

host = os.getenv('RM_HOST')
db_host = os.getenv('DB_HOST')
db_user = os.getenv('DB_USER')
db_pass = os.getenv('DB_PASSWORD')
database = os.getenv('DB_DATABASE')
username = os.getenv('RM_USER')
password = os.getenv('RM_PASSWORD')
phoneNumberList = ""
mailAddressList = ""

# Подключаем логирование
logging.basicConfig(
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn10Quit
```

Параметры БД задаются compose-файлом, отдельных изменений не требуется (.sql переложил).

ansible

Три стенда - первый уже использовавшийся ранее для ansible, второй (новый) для бота и primary базы, третий (новый) для replica базы:

The image displays a Kali Linux desktop environment with three terminal windows. The top window, titled 'user@devops-local: ~', shows the output of the 'neofetch' command, which displays system information such as OS (Debian GNU/Linux 12), Kernel (6.1.0-25-amd64), and a colorful ASCII art logo. The bottom-left window, titled 'user@devops-local-4: ~', shows the output of the 'hostnamectl' command for 'devops-local-4', displaying details like Machine ID, Boot ID, and Operating System. The bottom-right window, titled 'user@devops-local-5: ~', shows the output of the 'hostnamectl' command for 'devops-local-5', displaying similar system details. The desktop background is a dark purple gradient with a subtle grid pattern.

На новых шаблонных машинах уже есть git, python и pip без зависимостей. На рабочей плюс ansible:

The image shows two terminal windows side-by-side. The left window is titled 'user@devops-local-4: ~' and shows the following commands and output:
\$ python3 --version
Python 3.11.2
\$ pip3 --version
pip 23.0.1 from /usr/lib/python3/dist-packages/pip (python 3.11)
\$ git --version
git version 2.39.2
\$
The right window is titled 'user@devops-local: ~' and shows:
\$ python3 --version
Python 3.11.2
\$ pip3 --version
pip 23.0.1 from /usr/lib/python3/dist-packages/pip (python 3.11)
\$ ansible --version
ansible [core 2.14.16]
 config file = None
 configured module search path = ['/home/user/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
 ansible python module location = /usr/lib/python3/dist-packages/ansible
 ansible collection location = /home/user/.ansible/collections:/usr/share/ansible/collections
 executable location = /usr/bin/ansible
 python version = 3.11.2 (main, Aug 26 2024, 07:20:54) [GCC 12.2.0] (/usr/bin/python3)
 jinja version = 3.1.2
 libyaml = True
\$ git --version
git version 2.39.2
\$

По наработкам из прошлых заданий расписываем inventory и cfg файлы - все исходники в гите:

The image shows two terminal windows side-by-side. The left window is titled 'user@devops-local: ~/block-4-gh/block4' and shows the contents of the inventory file:
/home/user/bl-ck4/inventory [----] 0 L: [1+ 8 9/ 9] *(193 / 193b) <EOF> [*][X]
[bot]
192.168.11.138 ansible_user=user ansible_password=user

[primary]
192.168.11.138 ansible_user=user ansible_password=user

[replica]
192.168.11.139 ansible_user=user ansible_password=user
\$
The right window is titled 'mc [user@devops-local]: ~/block-4-gh/block4' and shows the contents of the ansible.cfg file:
/home/user/bl-4/ansible.cfg [----] 0 L: [1+ 3 4/ 4] *(61 / 61b) <EOF> [*][X]
[defaults]
inventory = ./inventory
host_key_checking = false
\$

Конфигурации для настройки БД и репликации так же из прошлых заданий (обновил ip для работы на новых серверах).

The image displays two terminal windows side-by-side, showing the configuration of the PostgreSQL `pg_hba.conf` file. The left window shows the original configuration, and the right window shows the modified configuration after adding two new host entries.

Left Window (Original Configuration):

```

mc [user@devops-local:~]-block4-gh/block4
user@devops-local:~-block4-gh/block4
/home/user/bl-tgresql.conf [----] 0 L: 57+36 93/817 *(3954/29804b) 0035 0x023 [*] [X]

# - Connection Settings -

listen_addresses = 'localhost, 192.168.11.138'<-----# what IP address(es) to listen on
<-----# comma-separated list of addresses;
<-----# defaults to 'localhost'; use '*' for all
<-----# (change requires restart)
port = 5432<-----# (change requires restart)
max_connections = 100<-----# (change requires restart)
#superuser_reserved_connections = 3<-----# (change requires restart)
unix_socket_directories = '/var/run/postgresql'<-----# comma-separated list of directories
<-----# (change requires restart)
#unix_socket_group = ''<-----# (change requires restart)
#unix_socket_permissions = 0777<-----# begin with 0 to use octal notation
<-----# (change requires restart)
#bonjour = off<-----# advertise server via Bonjour
<-----# (change requires restart)
#bonjour_name = ''<-----# defaults to the computer name
<-----# (change requires restart)

# - TCP settings -
# see "man tcp" for details

#tcp_keepalives_idle = 0<-----# TCP_KEEPIDL, in seconds;
<-----# 0 selects the system default
#tcp_keepalives_interval = 0<-----# TCP_KEEPINTVL, in seconds;
<-----# 0 selects the system default
#tcp_keepalives_count = 0<-----# TCP_KEEPCNT;
<-----# 0 selects the system default
#tcp_user_timeout = 0<-----# TCP_USER_TIMEOUT, in milliseconds;
<-----# 0 selects the system default

#client_connection_check_interval = 0<-----# time between checks for client
<-----# disconnection while running queries;
<-----# 0 for never

# - Authentication -
1help 2save 3mark 4replac 5copy 6move 7search 8delete 9pullDn 10quit

```

Right Window (Modified Configuration):

```

mc [user@devops-local:~]-block4-gh/block4
user@devops-local:~-block4-gh/block4
/home/user/bl-ql-repl.conf [----] 0 L: 57+36 93/817 *(3954/29785b) 0035 0x023 [*] [X]

# - Connection Settings -

listen_addresses = 'localhost, 192.168.11.139'<-----# what IP address(es) to listen on
<-----# comma-separated list of addresses;
<-----# defaults to 'localhost'; use '*' for all
<-----# (change requires restart)
port = 5432<-----# (change requires restart)
max_connections = 100<-----# (change requires restart)
#superuser_reserved_connections = 3<-----# (change requires restart)
unix_socket_directories = '/var/run/postgresql'<-----# comma-separated list of directories
<-----# (change requires restart)
#unix_socket_group = ''<-----# (change requires restart)
#unix_socket_permissions = 0777<-----# begin with 0 to use octal notation
<-----# (change requires restart)
#bonjour = off<-----# advertise server via Bonjour
<-----# (change requires restart)
#bonjour_name = ''<-----# defaults to the computer name
<-----# (change requires restart)

# - TCP settings -
# see "man tcp" for details

#tcp_keepalives_idle = 0<-----# TCP_KEEPIDL, in seconds;
<-----# 0 selects the system default
#tcp_keepalives_interval = 0<-----# TCP_KEEPINTVL, in seconds;
<-----# 0 selects the system default
#tcp_keepalives_count = 0<-----# TCP_KEEPCNT;
<-----# 0 selects the system default
#tcp_user_timeout = 0<-----# TCP_USER_TIMEOUT, in milliseconds;
<-----# 0 selects the system default

#client_connection_check_interval = 0<-----# time between checks for client
<-----# disconnection while running queries;
<-----# 0 for never

# - Authentication -
1help 2save 3mark 4replac 5copy 6move 7search 8delete 9pullDn 10quit

```

```
mc [user@devops-local]:~/block-4-gh/block4
/home/user/bl~4/pg_hba.conf [----] 34 L: [ 71+36 107/107] *(5189/5240b) 0032 0x020 [*][X]
#
# Put your actual configuration here
# -----
#
# If you want to allow non-local connections, you need to add more
# "host" records. In that case you will also need to make PostgreSQL
# listen on a non-local interface via the listen_addresses
# configuration parameter, or via the -i or -h command line switches.


# DO NOT DISABLE!
# If you change this first entry you will need to make sure that the
# database superuser can access the database using some other method.
# Noninteractive access to all databases is required during automatic
# maintenance (custom daily cronjobs, replication, and similar tasks).
#
# Database administrative login by Unix domain socket
local all postgres peer

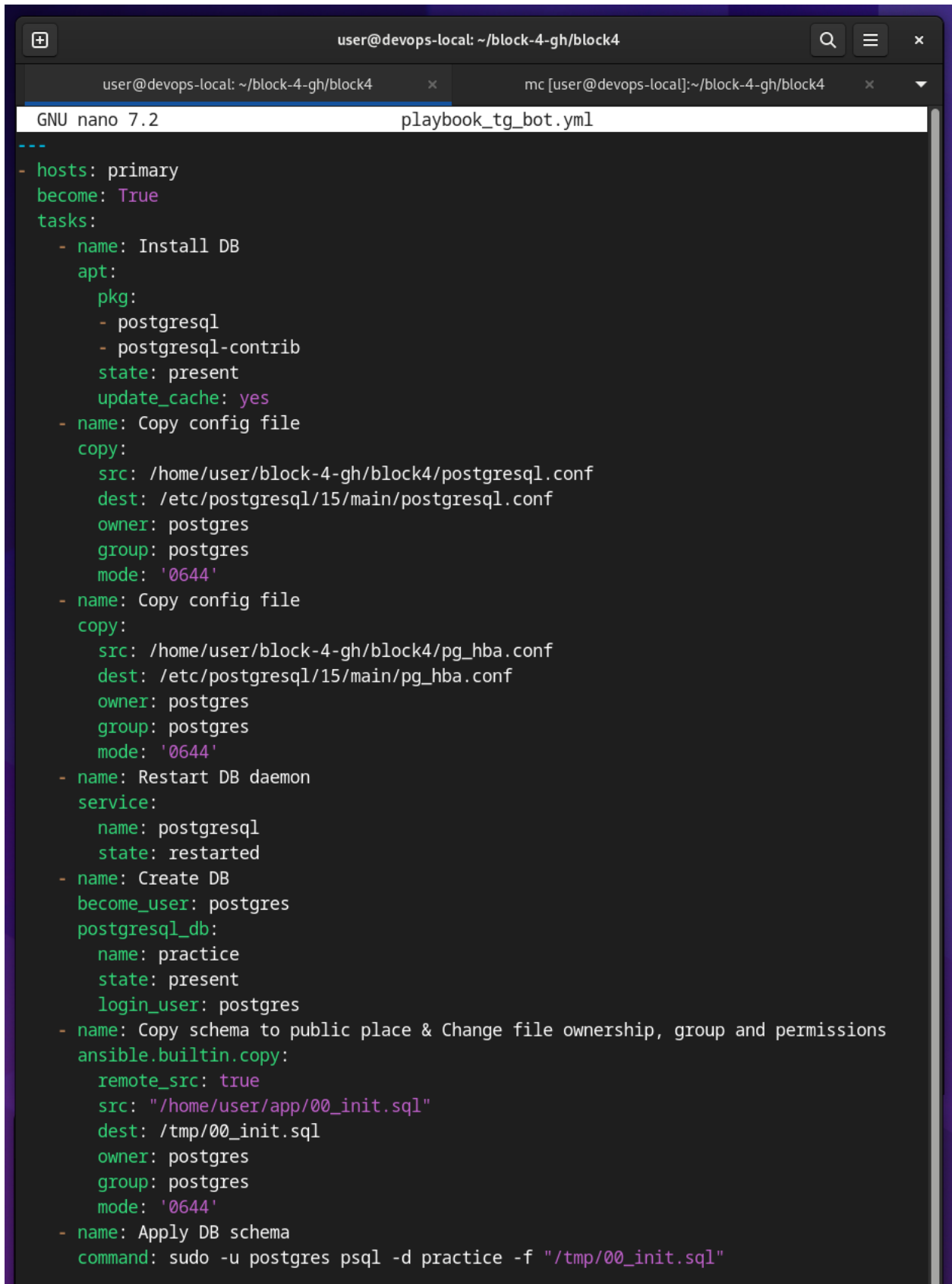
# TYPE DATABASE USER ADDRESS METHOD
# "local" is for Unix domain socket connections only
local all all peer
local all postgres trust
# IPv4 local connections:
host all all 127.0.0.1/32 scram-sha-256
host all all 192.168.11.138/24 scram-sha-256
# IPv6 local connections:
host all all ::1/128 scram-sha-256
# Allow replication connections from localhost, by a user with the
# replication privilege.
local replication all peer
host replication all 127.0.0.1/32 scram-sha-256
host replication all ::1/128 scram-sha-256
host replication replicator 192.168.11.139/24 scram-sha-256
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Плейбук состоит из двух задач - разворачивание БД с репликацией и бота.

Primary БД:

- С повышенными привилегиями устанавливаем необходимые приложения;
- Перекидываем конфиг БД и репликации;
- Ребутиим демона;

- Создание БД и применение ранее использованной схемы.



```
GNU nano 7.2          playbook_tg_bot.yml
---
- hosts: primary
  become: True
  tasks:
    - name: Install DB
      apt:
        pkg:
          - postgresql
          - postgresql-contrib
        state: present
        update_cache: yes
    - name: Copy config file
      copy:
        src: /home/user/block-4-gh/block4/postgresql.conf
        dest: /etc/postgresql/15/main/postgresql.conf
        owner: postgres
        group: postgres
        mode: '0644'
    - name: Copy config file
      copy:
        src: /home/user/block-4-gh/block4/pg_hba.conf
        dest: /etc/postgresql/15/main/pg_hba.conf
        owner: postgres
        group: postgres
        mode: '0644'
    - name: Restart DB daemon
      service:
        name: postgresql
        state: restarted
    - name: Create DB
      become_user: postgres
      postgresql_db:
        name: practice
        state: present
        login_user: postgres
    - name: Copy schema to public place & Change file ownership, group and permissions
      ansible.builtin.copy:
        remote_src: true
        src: "/home/user/app/00_init.sql"
        dest: /tmp/00_init.sql
        owner: postgres
        group: postgres
        mode: '0644'
    - name: Apply DB schema
      command: sudo -u postgres psql -d practice -f "/tmp/00_init.sql"
```

Для реплики примерно то же самое:

- Вместо создания БД - очистка имеющихся данных на случай перезапуска;
- Создание резервной копий работающего кластера - отправной точки ведомого сервера для потоковой репликации.

```
user@devops-local: ~/block-4-gh/block4
GNU nano 7.2                                playbook_tg_bot.yml

- hosts: replica
  become: True
  tasks:
    - name: Install DB
      apt:
        pkg:
          - postgresql
          - postgresql-contrib
        state: present
        update_cache: yes
    - name: Copy config file
      copy:
        src: /home/user/block-4-gh/block4/postgresql-repl.conf
        dest: /etc/postgresql/15/main/postgresql.conf
        owner: postgres
        group: postgres
        mode: '0644'
    - name: Stop DB daemon
      service:
        name: postgresql
        state: stopped
    - name: Clear data
      file:
        path: /var/lib/postgresql/15/main/
        state: absent
    - name: Start DB daemon
      service:
        name: postgresql
        state: started
    - name: Take a base backup of a PostgreSQL cluster
      become: yes
      become_user: postgres
      command: pg_basebackup -R -h 192.168.11.138 -U replicator -D /var/lib/postgresql/15>
      environment:
        PGPASSWORD: "replicator_password"
      register: pg_basebackup_result
      failed_when: pg_basebackup_result.rc != 0
      changed_when: False
```

Для бота:

- Клонирование гит-репы на сервер;
- Установка необходимых для работы бота библиотек с учетом окружения;
- Копирование переменных сред;

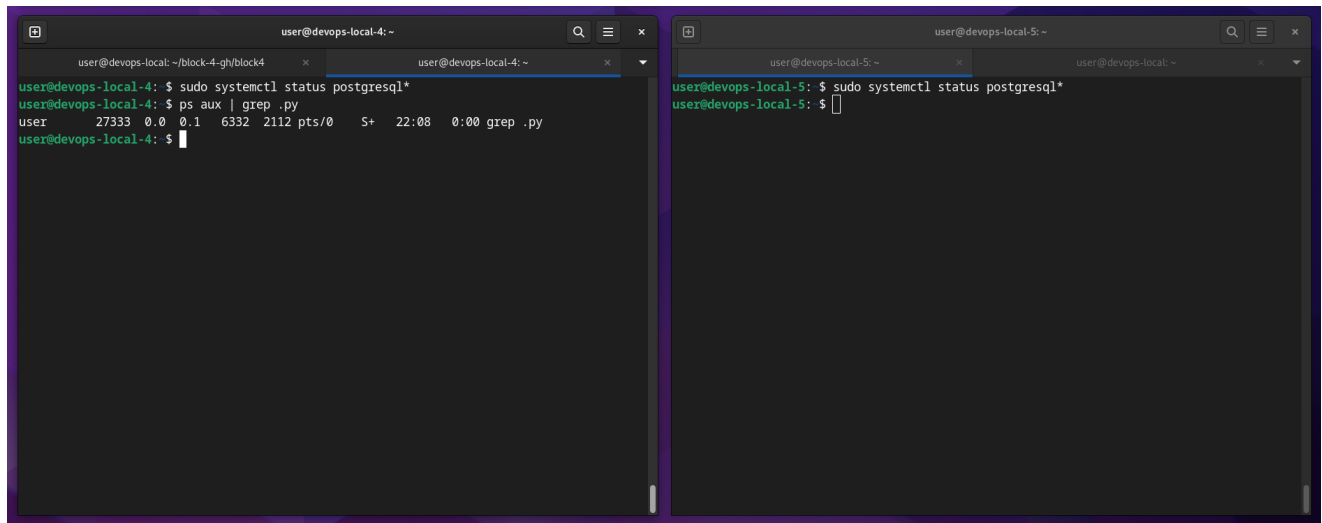
- Запуск бота в фоне с учетом особенностей использования ansible.

```
user@devops-local: ~/block-4-gh/block4
GNU nano 7.2      playbook_tg_bot.yml

- hosts: bot
  tasks:
    - name: Clone of a single branch
      git:
        repo: https://github.com/pkilipko/block4.git
        dest: /home/user/app
        version: ansible
        update: yes
    - name: Install specified python requirements
      pip:
        requirements: /home/user/app/requirements.txt
        virtualenv: /home/user/venv
        virtualenv_python: python3.11
        state: present
    - name: Copy file with permissions
      copy:
        src: /home/user/block-4-gh/block4/.env
        dest: /home/user/app/.env
        owner: user
        group: user
        mode: '0644'
    - name: Start script in background with Asynchronous actions and polling
      command: nohup /home/user/venv/bin/python3 block-2.py &
      args:
        chdir: "/home/user/app/"
      async: 9999
      poll: 0

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Фиксируем отсутствие на серверах postgres и запущенного бота:



The image shows two terminal windows side-by-side. The left window is titled 'user@devops-local-4: ~' and shows the command 'sudo systemctl status postgresql*' being executed. The output shows that PostgreSQL is not running. The right window is titled 'user@devops-local-5: ~' and shows the same command being executed, also resulting in PostgreSQL not being running.

```
user@devops-local-4: ~$ sudo systemctl status postgresql*
user@devops-local-4: ~$ ps aux | grep .py
user      27333  0.0  0.1  6332  2112 pts/0    S+   22:08   0:00  grep .py
user@devops-local-4: ~$

user@devops-local-5: ~$ sudo systemctl status postgresql*
user@devops-local-5: ~$
```

Запускаем на ansible-хосте плейбук - отмечаем, что все ключевые задачи принесли изменения на хосты:



The image shows a terminal window titled 'user@devops-local: ~/block-4-gh/block4'. It displays the output of an Ansible playbook run. The output shows various tasks being executed on two hosts: 192.168.11.138 and 192.168.11.139. The tasks include gathering facts, cloning a branch, installing python requirements, copying files, starting a script, installing a database, creating a database, copying a schema, applying a schema, and starting a database daemon. The output also shows a recap of the results for each host.

```
user@devops-local: ~/block-4-gh/block4$ ansible-playbook playbook_tg_bot.yml

PLAY [bot] *****
TASK [Gathering Facts] *****
ok: [192.168.11.138]

TASK [Clone of a single branch] *****
changed: [192.168.11.138]

TASK [Install specified python requirements] *****
changed: [192.168.11.138]

TASK [Copy file with permissions] *****
changed: [192.168.11.138]

TASK [Start script in background with Asynchronous actions and polling] *****
changed: [192.168.11.138]

PLAY [primary] *****
TASK [Gathering Facts] *****
ok: [192.168.11.138]

TASK [Install DB] *****
changed: [192.168.11.138]

TASK [Copy config file] *****
changed: [192.168.11.138]

TASK [Copy config file] *****
changed: [192.168.11.138]

TASK [Restart DB daemon] *****
changed: [192.168.11.138]

TASK [Create DB] *****
changed: [192.168.11.138]

TASK [Copy schema to public place & Change file ownership, group and permissions] *****
ok: [192.168.11.138]

TASK [Apply DB schema] *****
changed: [192.168.11.138]

PLAY [replica] *****
TASK [Gathering Facts] *****
ok: [192.168.11.139]

TASK [Install DB] *****
changed: [192.168.11.139]

TASK [Copy config file] *****
changed: [192.168.11.139]

TASK [Stop DB daemon] *****
changed: [192.168.11.139]

TASK [Clear data] *****
changed: [192.168.11.139]

TASK [Start DB daemon] *****
changed: [192.168.11.139]

TASK [Take a base backup of a PostgreSQL cluster] *****
ok: [192.168.11.139]

PLAY RECAP *****
192.168.11.138      : ok=13   changed=10   unreachable=0   failed=0   skipped=0   rescued=0   ignored=0
192.168.11.139      : ok=7    changed=5    unreachable=0   failed=0   skipped=0   rescued=0   ignored=0

user@devops-local: ~/block-4-gh/block4$
```

Отмечаем наличие новых служб postgresql на серверах и фонового процесса от бота:

```
user@devops-local-4: ~
CPU: 2ms
Sep 10 23:16:29 devops-local-4 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS.
Sep 10 23:16:29 devops-local-4 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.

• postgresql@15-main.service - PostgreSQL Cluster 15-main
  Loaded: loaded (/lib/systemd/system/postgresql@15-main.service; enabled-runtime; preset: en
  Active: active (running) since Tue 2024-09-10 23:16:29 MSK; 14s ago
  Process: 32564 ExecStart=/usr/bin/pg_ctlcluster --skip-systemctl-redirect 15-main star
  Main PID: 32570 (postgres)
  Tasks: 8 (limit: 2244)
  Memory: 20.1M
  CPU: 709ms
  CGroup: /system.slice/system-postgresql.slice/postgresql@15-main.service
          └─32570 /usr/lib/postgresql/15/bin/postgres -D /var/lib/postgresql/15/main
              └─32571 "postgres: 15/main: checkpointer "
                  └─32572 "postgres: 15/main: background writer "
                      └─32574 "postgres: 15/main: walwriter "
                          └─32575 "postgres: 15/main: autovacuum launcher "
```

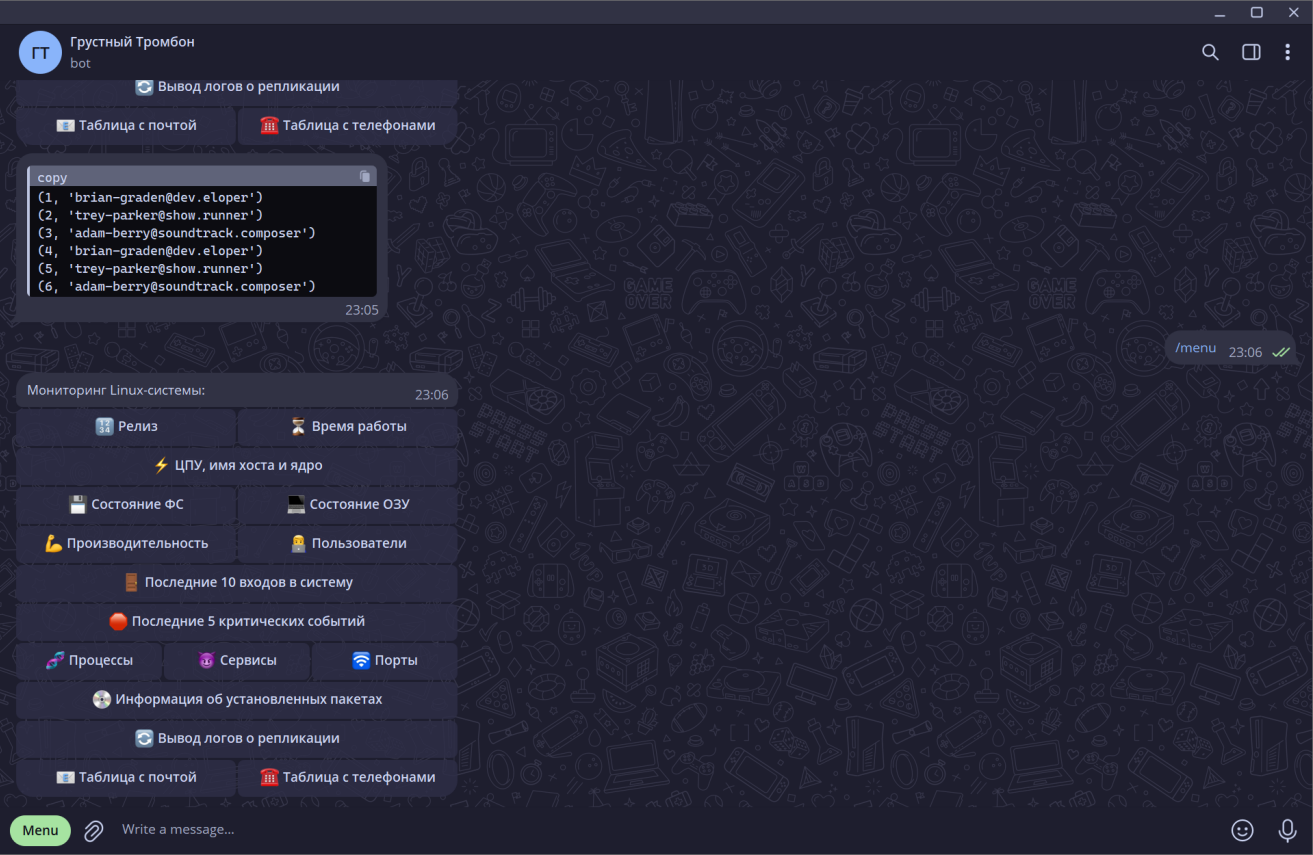
```
user@devops-local-4: $ ps aux | grep .py
user      32504  0.7  3.1 608700 62312 pts/1    Sl   23:14   0:01 /home/user/venv/bin/pyt
hon3 block-2.py
user      32603  0.0  0.1  6332  2168 pts/0    S+   23:16   0:00 grep .py
user@devops-local-4: $
```

```
user@devops-local-5: ~
• postgresql@15-main.service - PostgreSQL Cluster 15-main
  Loaded: loaded (/lib/systemd/system/postgresql@15-main.service; enabled-runtime; preset: en
  Active: active (running) since Tue 2024-09-10 23:16:37 MSK; 14s ago
  Process: 3294 ExecStart=/usr/bin/pg_ctlcluster --skip-systemctl-redirect 15-main star
  Main PID: 3301 (postgres)
  Tasks: 5 (limit: 2244)
  Memory: 17.8M
  CPU: 549ms
  CGroup: /system.slice/system-postgresql.slice/postgresql@15-main.service
          └─3301 /usr/lib/postgresql/15/bin/postgres -D /var/lib/postgresql/15/main
              └─3302 "postgres: 15/main: checkpointer "
                  └─3303 "postgres: 15/main: background writer "
                      └─3304 "postgres: 15/main: startup recovering 00000001000000000000000004"
                          └─3305 "postgres: 15/main: walreceiver streaming 0/4000008"

Sep 10 23:16:35 devops-local-5 systemd[1]: Starting postgresql@15-main.service - Postgres
Sep 10 23:16:37 devops-local-5 systemd[1]: Started postgresql@15-main.service - Postgres

• postgresql.service - PostgreSQL RDBMS
  Loaded: loaded (/lib/systemd/system/postgresql.service; enabled; preset: enabled)
  Active: active (exited) since Tue 2024-09-10 23:16:37 MSK; 14s ago
  Process: 3316 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
  Main PID: 3316 (code=exited, status=0/SUCCESS)
  CPU: 3ms
```

В боте кнопки таблиц ничего не показывают - база пустая:



Делаем поиск адресов в сообщении и добавляем их в базу:

